An agent-built pipeline for in silico antisense oligonucleotide design, and what six adversarial reviewers found in it

A methodological report on *ABCA4* c.161-395G>A

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Implementation, pipeline execution and drafting were carried out by an AI agent (Claude, Anthropic) under author supervision. Section 2 states explicitly which decisions were the authors' and which were not.

Abstract

The deep-intronic variant ***ABCA4* c.161-395G>A** activates an aberrant pseudoexon (PE1b, 91 bp) and was the most prevalent deep-intronic allele in a reported Chinese Stargardt disease cohort [1]. No antisense oligonucleotide (ASO) has been published or patented against it. We report a reproducible seven-module computational pipeline for in silico design of splice-switching ASO candidates of **PMO** chemistry [11], built end to end by an AI agent over six days, and then submitted to a panel of six independent adversarial reviewers.

The pipeline reproduces three results with external anchors. First, the variant supplies the −2 consensus adenine of the donor motif over an intact GT dinucleotide (`GGG|GTAGGT` → `GAG|GTAGGT`), a model-free and threshold-free observation. Second, two independently trained neural predictors [6, 7] place their maximum at exactly +1 and −89 relative to the variant, implying a 91 bp pseudoexon — which reproduces, *position for position*, the boundaries that Corradi et al. [3] had already measured by midigene assay in 2023 and that Wang et al. [1] later confirmed in size by minigene. Third, on an independent published oligo-walk with measured efficacy [2] the same scoring places the five known-effective AONs at ranks 1, 3, 4, 5 and 6 of 32 (AUC = 0.974).

We report the adversarial review as a primary result rather than as a limitations paragraph. The panel returned *reject with invitation to major resubmission* and identified seven critical issues. Correcting them changed the conclusions: a strand-orientation bug in the melting-temperature calculation had propagated four modules upstream and generated a finding, a design record and a published conclusion; a negative control with deliberately wrong tissue reproduced the "three-predictor agreement" exactly, showing that agreement to be uninformative; and a claim repeated across six documents for four days had never been verified against the literature. Assembling this manuscript's reference list — the last step before submission — produced two more of the same kind: the source paper had been misattributed to the wrong first author for five days, and the pseudoexon coordinates the project treated as its own had been published in 2023.

No candidate has been synthesised or assayed. We argue that for agent-built research the adversarial review is not optional quality control but part of the method, and we quantify why.

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1 Motivation

Stargardt disease type 1 is the most common inherited macular dystrophy, caused by biallelic variants in *ABCA4*. A relevant fraction of pathogenic alleles are not coding variants but **deep-intronic variants** that activate cryptic splice sites, leading the spliceosome to treat an intronic stretch as an exon — a *pseudoexon* — and to introduce a premature termination codon.

The variant **c.161-395G>A** has been reported by three independent groups. Corradi et al. [3] first characterised it in 2023 by midigene assay in a European cohort, reporting 36 % residual wild-type RNA and the exact pseudoexon boundaries. Wang et al. [1] showed by minigene that it activates three pseudoexons (PE1b 91 bp, PE1c 255 bp, PE1d 265 bp), with 76.4 % of the resulting mRNA incorrectly spliced, and found it to be the most prevalent deep-intronic allele of their Chinese cohort. Huang et al. [4] recovered it again in a Taiwanese cohort. None of the three reports an antisense oligonucleotide against it.

Splice-switching ASOs bind the pre-mRNA by base complementarity and **sterically block** access to a splice site without degrading the transcript [19]. No published or patented ASO targets this variant — which motivates the work, but also means **there is no positive control** with which to validate a design pipeline end to end. Section 4 describes the partial substitute we used instead.

2 Provenance and division of judgement

We state this before the methods because it conditions how the rest should be read.

The implementation, pipeline execution and drafting were produced by an AI agent. The following were author decisions, not the agent's:

- **The chemistry.** PMO was chosen as a scope decision, against the retinal literature precedent (2′-MOE/PS), and documented with its costs.
- **Commissioning the adversarial review**, on the explicit reasoning that an agent cannot audit work it produced.
- **Challenging the agent's assertions.** Three of the highest-impact errors surfaced because the authors asked, not because the agent detected them — as did the fact that PMO force-field parameters do exist [12], after the agent had asserted they did not.

We consider this disclosure a methodological requirement rather than an acknowledgement.

3 Pipeline

Seven modules, each with its own reproducible runner and declared limitations (Table 1, Figure 1).

#	Module	Question	Result
1	Sequence	Where is the variant?	Coordinate confirmed by two routes [17]
2	Oligo-walk	Which windows?	381 candidates
3	Heuristic filter	Viable oligos?	381 → 276
4	Thermodynamics	Bind well, reachable?	276 → 78
5	Off-target	Resemble other genes?	78 annotated by severity
6	Neural splicing	Is the false site real?	Donor +1, acceptor −89
6b	ASO masking	Does each patch switch it off?	12 abolish the pseudoexon
7	Ranking	Which to synthesise first?	Pareto front: 3

Table 1: The seven modules and what each one answers. Every row has a regeneration command in the repository README.

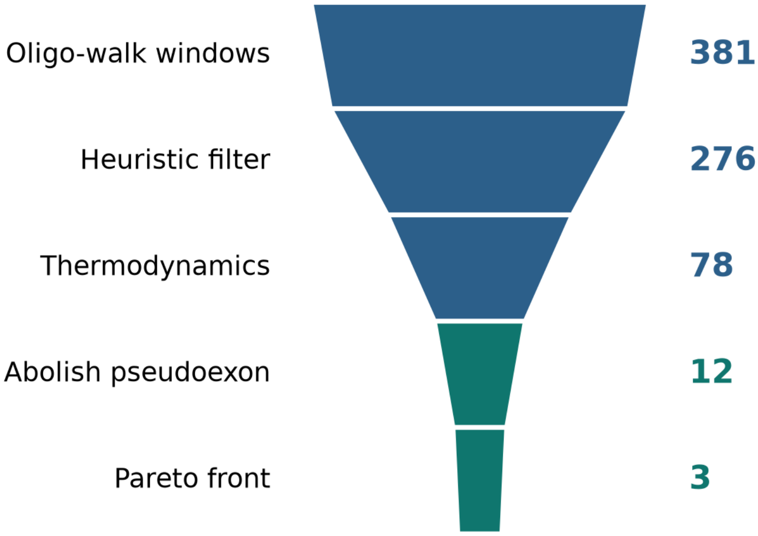

Figure 1: Candidate funnel with the current numbers. The two green stages are the ones that require neural inference; everything upstream is sequence arithmetic. The width of each block scales as $n^{0.42}$ so the last stages remain visible.

3.1 Representing an ASO to a splice predictor

Splice predictors take a single sequence strand and have no representation of a bound ligand. A PMO does not cleave or modify the RNA — it covers it. We therefore replace the ASO window with `N`, which the one-hot encoding of both predictor families maps to the null vector: "no readable sequence information here". This is a direct proxy for steric block.

Let x be the mutant pre-mRNA sequence, c a candidate window, and $\mathcal{M}(x, c)$ the same sequence with every base inside c replaced by `N`. For a predictor p and a splice site s, write $S_p(\cdot, s)$ for the score it assigns at that position. The quantity the pipeline actually uses is the *retained fraction of signal*:

$$r_p(c, s) \;=\; \frac{S_p\big(\mathcal{M}(x, c),\, s\big)}{S_p(x, s)} \tag{1}$$

$r_p = 1$ means the mask changed nothing; $r_p = 0$ means the site disappeared. Expressing the effect as a ratio rather than a difference is what makes the criterion comparable across predictors whose score scales differ.

A candidate is judged to **abolish the pseudoexon** when it switches off *at least one* of its two borders — the cryptic donor d at +1 or the cryptic acceptor a at −89 — in *every* predictor:

$$V(c) = \text{abolishes} \iff \forall p \in P : \min_{s \in \{d, a\}} r_p(c, s) \;\le\; \tau, \qquad \tau = 0.25 \tag{2}$$

The disjunction over borders is a biological statement, not a convenience: a pseudoexon needs both of its borders to be recognised, so removing either one is enough to disassemble it. The threshold τ is a declared design choice and is not calibrated against experiment.

Two alternatives were rejected: mutating the window introduces spurious motifs, and excising it shifts all downstream coordinates. Notably, the predictors' own inference routine already pads with `"N"`×5000 on each side, so blocks of `N` are the operational convention of these models rather than an invented input.

3.2 Thermodynamics, and the exact point where the chemistry does not fit

Module 4 estimates the melting temperature of the ASO:RNA duplex with the nearest-neighbour model [10], using the RNA/DNA hybrid table of Sugimoto et al. [9] as implemented in Biopython [16]:

$$T_m \;=\; \frac{\Delta H^\circ}{\Delta S^\circ + R\,\ln\!\big(C_T/x\big)} \;-\; 273.15 \tag{3}$$

with the enthalpy and entropy accumulated over consecutive dinucleotide steps of the sequence $\boldsymbol{s} = s_1 s_2 \ldots s_n$,

$$\Delta H^\circ = \Delta H^\circ_{\text{init}} + \sum_{i=1}^{n-1} \Delta H^\circ(s_i s_{i+1}), \qquad \Delta S^\circ = \Delta S^\circ_{\text{init}} + \sum_{i=1}^{n-1} \Delta S^\circ(s_i s_{i+1}). \tag{4}$$

The table is asymmetric, and that is what broke. For a hybrid duplex the parameter set is indexed by the **RNA** strand, so $\Delta H^\circ(\texttt{AG/TT}) \neq \Delta H^\circ(\texttt{TT/AG})$. In the tabulated values these differ by more than a factor of two (-3.96 vs -10.38 kcal mol^{-1}; -11.6 vs -31.8 cal mol^{-1}K^{-1}). Passing the ASO where the RNA target was expected therefore does not produce a small perturbation — it produces a different number. Section 6.1 reports what that cost.

Two biases that do not cancel. Equation 3 assumes a charged backbone, and the salt correction applied on top of it assumes ~50 mM Na$^+$. A PMO backbone is neutral [11], and published PMO melting measurements are made near 1.0 M NaCl. Independently, phosphorodiamidate linkages are reported to *stabilise* the duplex [12]. Both effects push the estimate in the same direction: the proxy **underestimates** the true melting temperature. This is why T_m is now reported as an annotation and not used as a filter (Section 6.4).

3.3 Off-target

Module 5 searches each candidate against the human transcriptome with BLAST+ [15] and records $L(c)$, the longest perfect contiguous match against any transcript other than *ABCA4*. Severity is annotated, not gated: for a steric-block chemistry, unlike for an RNase H-recruiting gapmer, homology only matters where it lands [19]. Two published results bound how much this measurement can be trusted: no single sequence feature predicts real off-target mis-splicing well, and BLAST itself penalises gaps and ignores G:U wobble pairing [18]; and the number of complementary sites grows steeply with the mismatch tolerance allowed [20].

3.4 Ranking: why a Pareto front and not a weighted score

The three ranking dimensions have incomparable units. All are written so that higher is better:

$$f_1(c) = 1 - \frac{1}{|P|}\sum_{p\in P} \min_{s\in\{d,a\}} r_p(c,s) \qquad \text{block strength} \tag{5}$$

$$f_2(c) = -\,L(c) \qquad \text{off-target safety} \tag{6}$$

$$f_3(c) = \tfrac{1}{2}\big(P_{\text{acc}}(c) + P_{\text{hd}}(c)\big) \qquad \text{thermodynamic quality} \tag{7}$$

where P_{acc} and P_{hd} are the within-cohort percentiles of target-window accessibility and of homodimer avoidance, computed with ViennaRNA [14].

Averaging f_1, f_2 and f_3 into one score requires positing an exchange rate — how many base pairs of off-target homology are worth 0.01 of block strength — that **no data in this project supports**, precisely because there is no positive control against which to calibrate it. A Pareto front requires no such rate. Writing $\boldsymbol{f}(c) = (f_1, f_2, f_3)$, candidate a *dominates* b when

$$a \succ b \iff \big(\forall i:\ f_i(a) \geq f_i(b)\big) \ \wedge\ \big(\exists j:\ f_j(a) > f_j(b)\big), \tag{8}$$

and the front is the set of candidates no one dominates, restricted to those that pass the eligibility gate $V(c) = \text{abolishes}$ of Equation 2:

$$\mathcal{F} \ = \ \big\{\, a \in E \ :\ \nexists b \in E,\ b \succ a \,\big\}, \qquad E = \{\, c : V(c) = \text{abolishes}\,\}. \tag{9}$$

It returns a set rather than a winner, which is the honest output.

The convention we have to declare. Collapsing accessibility and homodimer avoidance into a single average in Equation 7 *is* an implicit 50/50 weighting — exactly what this section claims to avoid. We do it anyway, for a measured rather than aesthetic reason: kept as separate dimensions, the front grows from 3 to **11 of the 12** eligible candidates, and a front that discards almost nothing informs almost nothing. The code reports this number so the decision stays visible instead of buried.

4 Calibration against an independent oligo-walk

Since no ASO exists for this variant, the pipeline was applied unchanged to a different *ABCA4* variant that *does* have measured efficacy: c.5461-10T>C in intron 38, for which Kaltak et al. [2] published a 32-AON walk and reported five as effective, one of which became the clinical candidate QR-1011. The calibration module is deliberately parallel to the main pipeline and does not touch it.

Scoring the 32 AONs by the same masking Δ used in Section 3.1 and asking where the known-effective ones land gives a Mann–Whitney statistic. With K the known-effective set and O the rest,

$$U = \sum_{k \in K} \sum_{o \in O} \Big[\mathbf{1}\big(\delta_k > \delta_o\big) + \tfrac{1}{2}\,\mathbf{1}\big(\delta_k = \delta_o\big)\Big], \qquad \text{AUC} = \frac{U}{|K|\,|O|}. \tag{10}$$

The five known-effective AONs occupy ranks **1, 3, 4, 5 and 6 of 32**, with QR-1011 third: AUC = 0.974. Repeating the whole calibration with the retina-trained predictor [8] gives the same five ranks and AUC = 0.970, so the separation is not an artefact of one model. A sensitivity analysis removing the four AONs that cover the exon 39 acceptor — where abolishing the site is trivially bad — leaves 28 AONs and AUC > 0.95.

What this does and does not license. It shows the scoring is not arbitrary: it discriminates AONs with measured efficacy on a target it was never tuned for. It does not transfer to c.161-395G>A, which is a different intron, a different mechanism (pseudoexon inclusion rather than exon skipping) and a different chemistry.

5 Results with external anchors

These two do not depend on any metric we defined.

5.1 The variant supplies the donor consensus adenine

Counting matches against the donor consensus `MAG|GTRAGT`, the wild-type sequence 1 nt downstream of the variant reads `GGG|GTAGGT` and the mutant reads `GAG|GTAGGT`. The variant itself supplies the adenine at position −2 — one of the most conserved positions of the donor consensus — over an intact GT dinucleotide, with purines at +3 and +5. This requires no model, no threshold and no predictor.

5.2 The predicted pseudoexon reproduces published coordinates, position for position

SpliceAI [6] local inference over a 10 kb window predicts a cryptic donor strengthened at +1 (0.194 $\rightarrow$ 0.560) and a cryptic acceptor at −89 (0.061 $\rightarrow$ 0.271), implying a pseudoexon of **91 bp**.

Corradi et al. [3] measured this variant with a midigene splice assay in 2023 and reported the resulting transcript as `r.160_161ins161-484_161-394`. Converted to offsets relative to the variant at `c.161-395`, that published RNA variant states the acceptor is at $395 - 484 = -89$, the donor at $395 - 394 = +1$, and the pseudoexon is $484 - 394 + 1 = 91$ nt. **The prediction matches the published experiment at both boundaries, not merely in length.** Wang et al. [1] independently confirmed the 91 bp size by minigene, as PE1b.

A third cohort provides a second kind of replication. Huang et al. [4] report SpliceAI delta scores of DS_AG = 0.21 and DS_DG = 0.37 for this variant; our local inference gives $0.271 - 0.061 = 0.210$ and $0.560 - 0.194 = 0.366$. Different group, different pipeline, different cohort, same deltas to two decimals.

Pangolin [7], a different architecture trained by a different group, places its maximum at *exactly* the same two positions, with neighbouring positions at ∼1e−5. A deliberate attempt to break this result during adversarial review failed.

This is a validation, not a discovery. We state plainly that the coordinates were already in the literature: Corradi et al. published them two years before this project began, and the project did not know it until the reference list for this preprint was assembled (Section 6.6). What the agreement establishes is that the pipeline recovers a functionally measured pseudoexon from sequence alone — which is what a design pipeline needs to be trusted for, and which is worth more here than a novelty claim would have been.

The variant strengthens a site that already exists. Our own wild-type donor score is 0.194, which is not zero, and Huang et al. [4] queried SpliceVault and found that both cryptic sites are used in normal transcriptomes, below 3 % of transcripts. So the correct statement is that the base change *reinforces* a weak pre-existing cryptic site — consistent with the consensus-adenine observation above — and not that it creates one *de novo*. Earlier drafts of this work said “creates”; that was too strong.

What does not follow. SpliceAI does not explain PE1c or PE1d: their implied acceptors score 0.016 and 0.042, far below any usable threshold. The model accounts for one of three reported pseudoexons, and we state this rather than omit it.

5.3 Which windows switch the pseudoexon off

Applying Equations 1 and 2 to the 78 candidates that reach Module 6b gives Figure 2. Two distinct mechanisms appear, and they are not interchangeable.

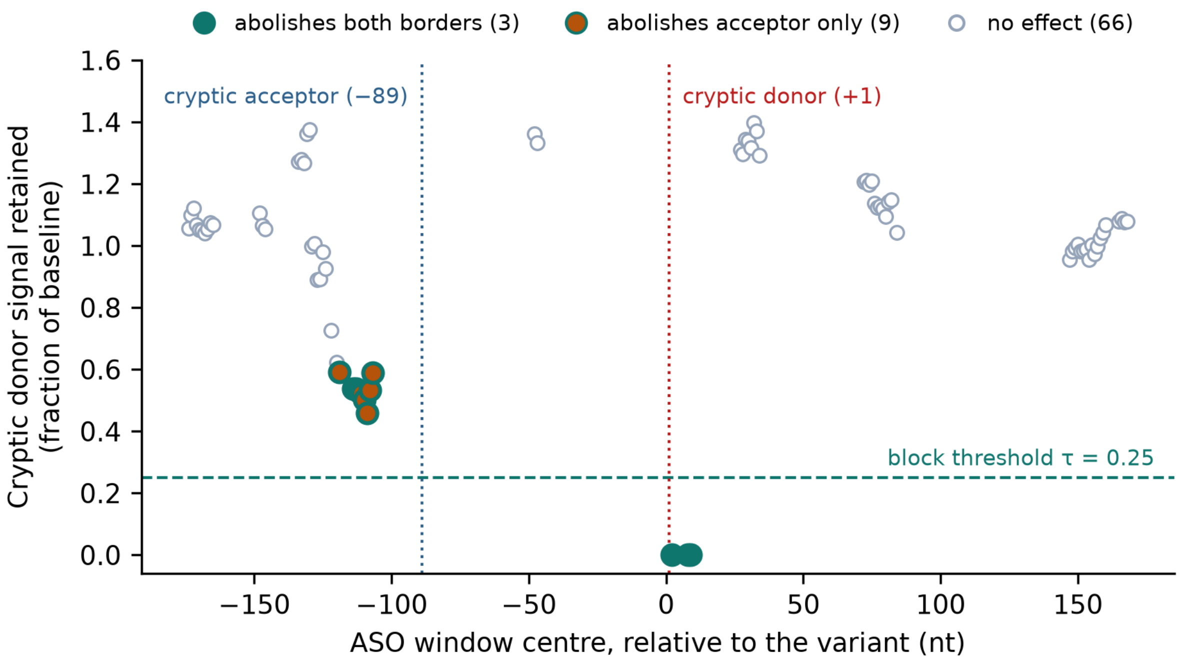

Figure 2: Effect of masking each candidate window on the cryptic donor signal, as a fraction of the unmasked baseline (Equation 1). Windows covering the donor at +1 drive it to zero. Nine candidates centred near −110 abolish the *acceptor* without covering the donor, and are visible here only as a partial drop in donor signal — their acceptor retention is what Equation 2 evaluates. The dashed line is the declared threshold $\tau = 0.25$.

6 The adversarial review as a result

Six independent reviewers — splicing biology, statistics, reproducibility, ASO therapeutics, code integrity, and a hostile red team — were given a dossier of every claim with its evidence level and regeneration command, plus explicit rules of engagement requiring that each objection cite a file, line or number.

The verdict was *reject with invitation to major resubmission*. Seven critical issues were raised. Four were independently verified before acceptance. Figure 3 summarises what changed; the subsections give the mechanism of each.

6.1 A Module 4 bug that travelled four modules upstream

`Bio.SeqUtils.MeltingTemp.Tm_NN` with the `R_DNA_NN1` table documents that the argument must be the **RNA** strand; the code passed the ASO. As shown in Section 3.2, the table is asymmetric, so the order changes the number.

Correcting it moved the Module 4 funnel from **44 to 16** candidates, with only 6 in common. More consequentially, the seven candidates that abolished the acceptor *without covering it* — the observation that had motivated the project's verdict criterion, its own design record, and a published conclusion about predictor agreement — **did not survive**. A defect in the cheapest module had generated a finding four modules downstream: not a wrong number, but a wrong claim about biology.

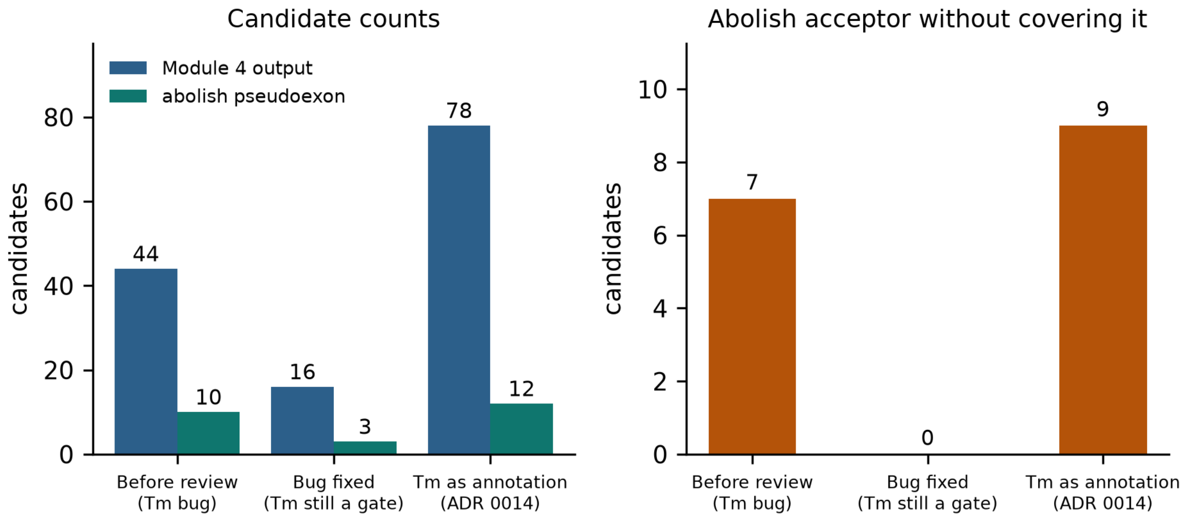

Figure 3: What the review changed. **Left:** candidates surviving Module 4 and, of those, how many abolish the pseudoexon. **Right:** candidates that abolish the cryptic acceptor without covering it — the observation that had motivated the project's verdict criterion. It existed before the review for the wrong reason (a bug), vanished entirely when the bug was fixed, and reappeared on solid ground once T_m stopped being used as a filter.

6.2 A wrong-tissue control that dissolved an argument

The project had reported that three independently trained predictors, one of them retina-specific [8], selected the same exact candidate set, and presented this as cross-validation. The panel proposed the missing control: run a model trained on *deliberately* the wrong tissue — the GTEx weight set released alongside Retina-SpliceAI, which excludes neural retina (Figure 4).

The wrong tissue selects the retina model's candidates one for one: both return the same 11 of 78, candidate for candidate, not merely the same count. The agreement therefore reflects shared architecture and shared training annotations, not tissue-robust biology. The control cost six minutes of CPU and had been available since the weights were installed; nobody in the project had run it, because nobody was looking for a way to refute their own argument.

6.3 A claim repeated for four days without verification

The assertion that no nearest-neighbour thermodynamic parameters exist for PMO propagated through six documents from a single unverified inference. Checking it found: the claim is essentially correct for PMO, but the NN model *is* documented as applicable to uncharged oligomers; parameters *do* exist for 2′-MOE/PS [13], the chemistry of the actual retinal literature; and PMO force-field parameters *do* exist [12]. Further, the experimental evidence shows phosphorodiamidate linkages *stabilise* the duplex, so the proxy **underestimates** the true melting temperature — a bias in the same direction as the inapplicable salt correction, so the two accumulate rather than cancel.

6.4 A correction that recovered what the filter had hidden

Because the melting-temperature threshold could not be interpreted for this chemistry, it was converted from a filter into an annotation — following a precedent the project had already set for off-target severity. The funnel went from 16 to **78** candidates.

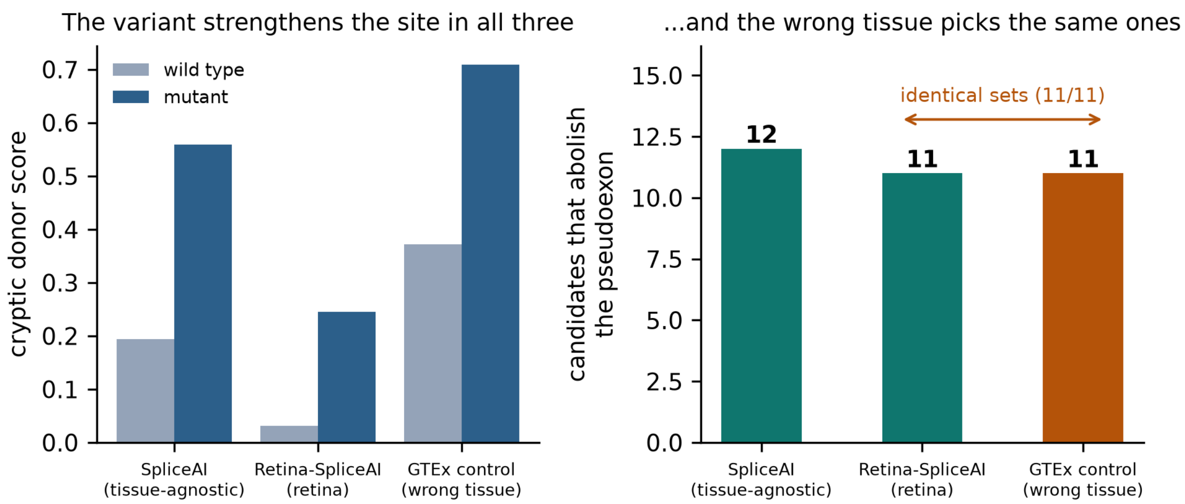

Figure 4: The negative control. **Left:** all three weight sets — tissue-agnostic, retina-trained, and a GTEx model that deliberately excludes neural retina — agree that the variant strengthens the cryptic donor, though they disagree on its absolute strength. **Right:** the retina model and the wrong-tissue model select *identical* candidate sets (11 of 78); the tissue-agnostic model differs by one. Since the wrong tissue reproduces the result, agreement between them cannot be evidence of tissue-robust biology.

The acceptor-only candidates **reappeared**: nine of them, with proxy melting temperatures of 36.5–40.1 °C, all below the discarded threshold. The mechanism was real; the buggy calculation had admitted them for the wrong reason and the corrected gate had hidden them entirely. The final Pareto front returns to the same three candidates as the original analysis, now by a defensible route.

6.5 A misattributed citation, found while assembling this bibliography

While compiling the reference list for this preprint, the source of the minigene measurement was checked against Crossref and PubMed Central for the first time. The DOI and PMC identifier the project had recorded were correct; the author name was not. The paper is **Wang et al.** [1] — there is no author named Peng on it. The attribution had been wrong in 19 places across the repository and in 15 vault documents, for five days, and had already been read by the six reviewers without any of them flagging it.

We report it because it is the same failure mode as the other three: a claim that was never checked against its source, repeated until repetition made it look verified. It is also the cheapest of the four to have caught — one API call — and, at the time, the last one found.

6.6 And the coordinates we thought were ours had been published in 2023

The same bibliography work produced a fifth instance, and the most consequential. The project had recorded that the source paper reports no cryptic site positions. That is true of Wang et al. [1], and was verified against their text. The project then generalised it to *nobody* having reported them — which was never checked, and is false.

A full-text search of the literature for the variant string returns exactly three papers. The earliest, Corradi et al. [3], measured the variant by midigene assay in September 2023 and published the pseudoexon with both of its boundaries, in the notation quoted in Section 5.2.

The third, Huang et al. [4], appeared in October 2025 with a Taiwanese cohort and reports the same SpliceAI deltas we compute.

Two things follow, and they point in opposite directions. The external anchor is **stronger** than we had claimed — position-for-position agreement with an independent functional assay rather than a coincidence of length. And the mechanistic contribution is **smaller**: none of the coordinates are new.

The instructive part is why the earlier check missed it. A web search had been run and returned nothing — but it searched for the *mechanism* of the variant rather than for the variant string in full text. The conclusion drawn at the time — "absence of a result is not proof of absence, so do not claim novelty" — was the right conclusion reached for the wrong reason: this was not irreducible uncertainty, it was a paper nobody had looked for correctly. That reflex to withhold the claim is the only thing that kept the error out of the published record. **The query matters as much as the source.**

One premise did survive the search, and is now checked rather than assumed: **no antisense oligonucleotide has been published against this variant**. Corradi et al. mention none. Sangermano et al. [5] — whose title promises AON correction of *ABCA4* splice defects, and which was therefore the most dangerous paper for this premise — designed AONs for six other variants and none for this one. Neither 2025 cohort reports AON work on it.

7 The candidates that survive

Twelve of the 78 candidates abolish the pseudoexon under Equation 2 in both predictors. Three are non-dominated under Equation 8 (Table 2).

Candidate	**Window** (rel. to variant)	**Borders** abolished	f_1 block	f_2 off-target	f_3 thermo
`cand_5992`	-8 to $+12$	donor + acceptor	1.000	-15	8.45
`cand_5998`	-2 to $+18$	donor + acceptor	1.000	-16	9.75
`cand_5882`	-118 to -98	acceptor only	0.994	-15	70.75

Table 2: The Pareto front. Each candidate is 20 nt. f_2 is the negated longest perfect contiguous match against another transcript (Equation 6), so -15 is safer than -16. Note the trade-off the front exposes: the two strongest blockers are the two *worst* on physicochemical properties, and the acceptor-only candidate is the reverse. Nothing in this project determines which trade-off is preferable — that is a human decision, which is why the pipeline returns three rows and not one.

No candidate has been synthesised, transfected or assayed. Table 2 is a synthesis order of priority, conditional on someone deciding to synthesise anything at all.

8 What this pipeline does not do

Stated with every result and not omissible when communicating it.

- **No experimental validation.** No ASO synthesised or assayed.
- **The masking proxy does not measure efficacy.** Equation 1 is binary and total: it assumes 100 % occupancy and perfect block. Necessary, not sufficient.
- **PMO thermodynamics are a proxy of another chemistry**, now correctly oriented but still not PMO, and biased in a known direction (Section 3.2).
- **Off-target severity thresholds are uncalibrated** design choices.

- **Predictor agreement is not independent** and, per the tissue control, largely uninformative.
- **The pipeline is blind to splice-site displacement**: it scores four fixed offsets and discards the rest of the profile.
- **The threshold $\tau = 0.25$ in Equation 2 is a declared convention**, not a measured cut-off.

9 Discussion

The scientific contribution of this work is modest and bounded, and one draft narrower than we first wrote it. The pipeline recovers, from sequence alone, a pseudoexon whose boundaries were established by functional assay in 2023 [3]; it does not discover them. What remains is a reproducible pipeline that reconstructs a measured splicing defect and produces a shortlist no one has tested.

The methodological contribution is, we think, larger. An AI agent produced a system with real engineering discipline — 224 tests, three documented environments, provenance files, archived pre-correction results — and simultaneously produced conclusions that did not survive contact with adversarial review. Crucially, **the failures were not arithmetic**. Six reviewers found no calculation error. They were failures of framing: which metric to define, which control not to run, which claim to repeat without checking — and, as Sections 6.5 and 6.6 show, which citation never to open and which search never to run.

Two of those failures surfaced only while assembling this reference list, after the adversarial review had concluded. That timing is itself a result: **writing the citations audited the project's premises more effectively than six specialised reviewers had**, because it was the first step that forced every external claim to be resolved against an identifier rather than against the project's own memory of it. It is also the cheapest step in the whole project.

That asymmetry is the finding we would emphasise. Agent-built research is likely to be strong where the work is mechanical and weak where it requires deciding what would refute you. Adversarial review is therefore not quality control appended at the end; for this mode of work it is part of the method.

10 Availability

Code, results and regeneration commands: `https://github.com/smnunez404/bio-oligonucleotidos` (MIT licence; generated data CC BY 4.0). Every number cited here has a regeneration command in the repository README, and the figures in this preprint are generated from `data/results/` by a versioned script, so text and figure cannot diverge. The adversarial review report and the claims dossier are maintained in the project's documentation vault and are available on request.

Acknowledgements

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